CS 215: Mock Exam

Name:	ID:

Notes:

Teaching-team notes.

1. When in doubt, state any valid assumptions and proceed. Answers without reasoning do not qualify for full credit. Write all answers on this question paper.
2. Use the (non-printed) back of the paper for rough work. These will not be used to award credit.
3. Anything written outside the box will not be evaluated.
4. You may assume any result stated in provided lecture handouts as "known" i.e., not to be proved. A specific instruction may override this, so read carefully.

1. (6mark) For independent, normal $X \sim \mathcal{N}(\mu_X, \sigma_X^2)$ and $Y \sim \mathcal{N}(\mu_Y, \sigma_Y^2)$, provide the mean and variance of the following if they are necessarily normal. Give one line reason (see the last note).

   A. $3X - 2$ B. $X + Y$ C. XY D. $|X|$ E. e^X F. $X - Y$